With or without replacement

1 A pile of playing cards has 4 diamonds and 3 hearts. A second pile has 2 diamonds and 5 hearts. One card is selected at random from the first pile, then the second.

a Construct a probability tree of this situation with the correct probability on each branch.

b Find the probability of selecting two hearts.

2 In tennis if the first serve is a fault (out or in the net), the player takes a second serve. A player serves with the following probabilities:

- First serve in: 0.55
- Second serve in: 0.81

a Construct a probability tree showing the probability of the first two serves either being in or a fault.

b Find the probability that the player needs to make a second serve.

c Find the probability that the player makes a double fault (both serves are a fault).

3 A container holds four counters coloured red, blue, green and yellow. Draw a tree diagram representing all possible outcomes when two draws are done, and the first counter is:

a Replaced before the second draw.

b Not replaced before the next draw.

4 A container holds three cards coloured red, blue and green.

a Draw a tree diagram representing all possible outcomes when two draws occur, and the card is not replaced before the next draw.

b Find the probability of drawing the blue card first.

c Find the probability of drawing a blue card in either the first or second draw.

d Find the probability of drawing at least one blue card if the cards are replaced before the next draw.

5 Consider the word WOLLONGONG. If three letters are randomly selected from it without replacement, find the probability that:

a The letters are W, O, L, in that order.

b The letters are O, N, G, in that order.

c All three letters are O.

d None of the three letters is an O.

6 Two marbles are randomly drawn without replacement from a bag containing 1 blue, 2 red and 3 yellow marbles.

a Construct a tree diagram to show the sample space.

b Find the probability of drawing the following:

i A blue marble and a yellow marble, in that order.

ii A red marble and a blue marble, in that order.

iii 2 red marbles.

iv No yellow marbles.

v 2 blue marbles.

vi A yellow marble and a red marble, in that order.

vii A yellow and a red marble, in any order.

7 There are 4 green counters and 8 purple counters in a bag. Find the probability of choosing a green counter, not replacing it, then choosing a purple counter.

8 A standard deck of cards is used and 3 cards are drawn out. Find the probability, in fraction form, of the following:

a All 3 cards are diamonds if the cards are drawn with replacement.

b All 3 cards are diamonds if the cards are drawn without replacement.

9 A number game uses a basket with 6 balls, all labelled with numbers from 1 to 6. 2 balls are drawn at random. Find the probability that the ball labelled 3 is picked once if the balls are drawn:

a With replacement.

b Without replacement.

10 A hand contains a 10, a jack, a queen, a king and an ace. Two cards are drawn from the hand at random, in succession and without replacement. Find the probability that:

a The ace is drawn.

b The king is not drawn.

c The queen is the second card drawn.

11 Eileen randomly selects two cards, with replacement, from a normal deck of cards. Find the probability that:

a The first card is a queen of spades and the second card is a 4 of clubs.

b The first card is spades and the second card is a 4.

c The first card is a Queen and the second card is black.

d The first card is not a 7 and the second card is not Clubs.

12 Find the probability of drawing a green co[redacted] from a bag of 5 green counters and 6 black counters, replacing it and drawing another green counter.

13 A chess player is placed into a draw where in each match he has a 30% chance of winning. Find the probability that:

a He wins his first two matches.

b He wins his first three matches.

c He wins his first two matches and then loses his third match.

14 Beth randomly selects three cards, with replacement, from a normal deck of cards. Find the probability that:

a The cards are queen of diamonds, king of spades, and king of diamonds, in that order.

b The cards are all black.

c The first card is a 9, the second card is a heart and the third card is red.

d The cards are all hearts.

e None of the cards is a 9.

15 From a standard pack of cards, one card is randomly drawn and then put back into the pack. A second card is then drawn. Calculate the probability that:

a Neither of the cards are diamonds

b At least one of the cards is a diamond

16 Three marbles are randomly drawn with replacement from a bag containing 6 red, 4 yellow, 3 white, 2 black and 4 green marbles. Find the probability of drawing:

a Three white marbles

b No green marbles

c At least one red marble

d At least one white marble

17 Amelia randomly selects two cards, with replacement, from a normal deck of cards. Find the probability that:

a Both cards are red

b Both cards are the same colour

c Both cards are of different colours

18 A bag contains four marbles - red, green, b[redacted]nd yellow. Beth randomly selects a marble, returns the marble to the bag and selects another marble.

a Construct a tree diagram for the experiment given.

b Find the probability of Beth selecting:

i A blue and a yellow marble.

ii A blue followed by a yellow marble.

iii 2 red marbles.

iv 2 marbles of the same colour.

v 2 marbles of different colours.

19 James, a test cricketer, analysed his past innings and found his probabilities of scoring particular numbers of runs, as shown in the table:

Number of runs	0	$1-20$	$21-49$	$50-99$	$100+$
Probability	$\frac{1}{10}$	$\frac{3}{10}$	$\frac{3}{10}$	$\frac{2}{10}$	$\frac{1}{10}$

If James is selected to play in the next test match, calculate the probability that he scores:

a 0 runs in exactly one of the two innings.

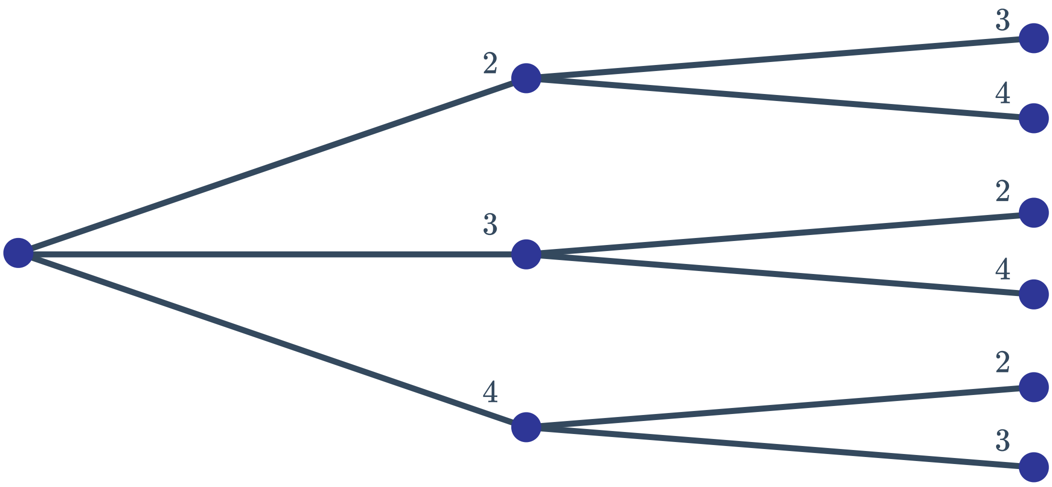
b 0 runs in both innings.

c At least 50 runs in at least one of the innings.

d A total of 100 or more runs in each of the two innings.

e More than 20 runs in only one of the two innings.

20 Three cards labeled 1, 2, 3 are placed face[redacted]n on a table. Two of the cards are selected randomly to form a two-digit number. The possible outcomes are displayed in the following probability tree:



a List the sample space of two digit numbers produced by this process.

b Find the probability that 2 is a digit in the number.

c Find the probability that the sum of the two selected cards is even.

d Find the probability of forming a number greater than 40.

21 In a school, 25% of students ride skateboards and 20% of students have dark hair. One student is selected at random. Find the probability that the student:

a Has dark hair and rides a skateboard.

b Has light hair and does not ride a skateboard.

c Has dark hair and does not ride a skateboard.

d Has light hair and rides a skateboard.

22 The ratio of left-handed people to right-handed people in a country is $4 : 3$. Two people are surveyed at random. Calculate the probability that:

a Both people are left-handed.

b One person is left-handed and the other is right-handed.

c At least one person is right-handed.